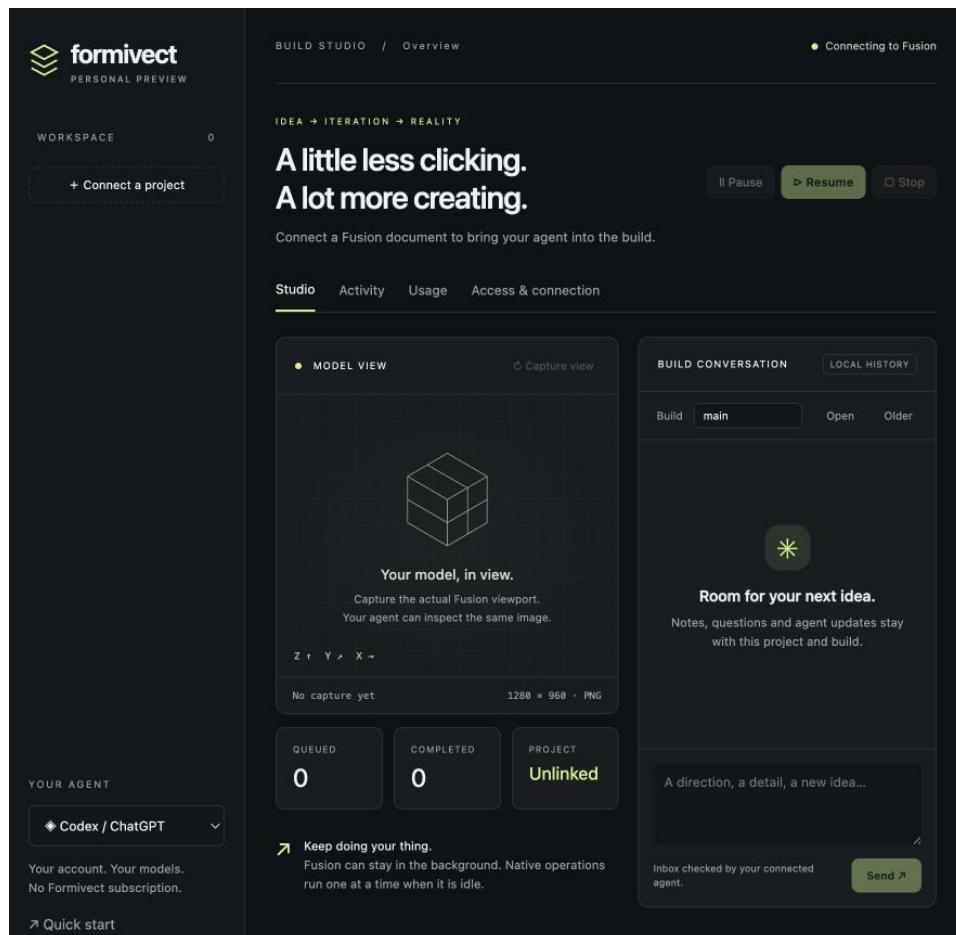


Formivect

Engineering AI Workspace

September 2026 / Personal preview 0.2.0

Working Windows Fusion preview. Independent platform proposed.

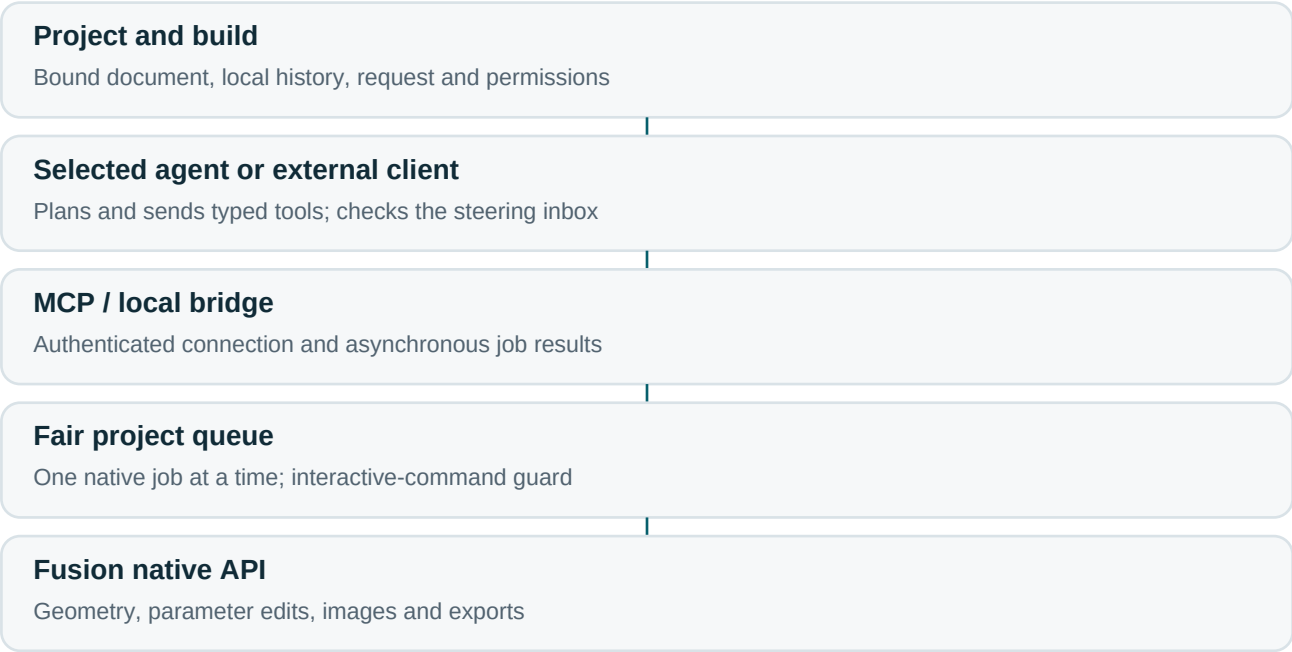


Original interface preview in its unconnected state; native execution evidence appears on page 2.

Formivect connects conversation to editable engineering work. The current add-in brings project-scoped AI tools, viewport access, local histories, operation controls and selected-model runs into Autodesk Fusion. The long-term aim is a standalone environment for CAD, analysis, manufacturing and an owned engineering model.

AI-assisted workflows supported the creation of this project.

A working native bridge



The palette connects an open document to a project, separates build conversations and lets the user select a provider/model/effort. Agents submit typed asynchronous operations; Fusion executes native jobs serially with permissions and document restoration.

Verified record	Scope
37 automated checks	Windows Python 3.14; source, installation and run lifecycle checks
17 MCP tools	11 native operation tools plus six status/job/control/message tools
11 native acceptance checks	In addition to two-design smoke, background and interactive-command tests
Selected-model runs	Actual Codex box inspection, viewport capture and visible build reply

The preview also supports reviewed local learning records and eligible dataset export. Claude/local launch adapters exist but remain unverified with real providers on the validation host. There is no trained Formivect engineering model.

One editable engineering project

Requirements + revision + selection

A proposed shared engineering project contract

Owned engineering model + retrieval

Typed actions, explanations and failure repair

Policy + durable execution

Revision guards, receipts, budgets and recovery

CAD / analysis / manufacturing workers

Exact engines and reproducible observations

Evidence + editable change set

Manual review, subsequent edits and checked outcomes

The proposed product owns its workspace, project/revision data, engineering execution and deployed model route. Users and agents operate the same editable feature history. Requirements link to geometry, analyses, drawings, BOMs, manufacturing plans and physical test evidence.

Potential features span parametric CAD, assembly motion, simulation, CAM, visual grounding, durable background work, coordinated agents, reusable libraries, revision comparison, change-impact detection and evaluated model specialization.

The model would propose typed operations. Geometry kernels, constraints, numerical solvers, manufacturing engines and independent checks would compute the results. The product is intended to remain useful as a project evolves through further manual and agent edits.

This page describes the proposed architecture. It is not delivered CAD-suite parity or a standalone application.

Build evidence before breadth

Windows native execution and one scoped advanced-script modeling workflow are demonstrated. The next stage is a useful recurring workflow with a fixed contract and an independently controlled evaluation suite.

Stage	Required outcome
1 / Choose a useful workflow	Confirm a representative recurring task and expert-reviewable acceptance contract. The mounting-plate change order remains a candidate.
2 / Freeze independent evaluation	Hold out design families; define evaluator-owned geometry and editability oracles, errors and a full manual baseline.
3 / Make tools and recovery reliable	Build typed feature/constraint operations, stable targets, revision guards, checkpoints and interruption recovery.
4 / Establish model and data baselines	Compare licensed candidates; curate permissioned, executed trajectories with checked labels and provenance.
5 / Evaluate specialization	Train versioned candidates and compare unseen-task success, correction time, latency and resource cost; retain rollback.
6 / Validate independent operation	Integrate native engines, manual editing, persistent execution and owned inference into reviewed real-user workflows.
7 / Expand supported engineering	Add analyses, manufacturing, drawings and broader CAD with separate benchmarks, platform tests and production support.

Current release position. The personal preview has no Formivect payment screen. Existing Fusion and AI access still apply; full commercial pricing, support and software distribution remain undecided.

This selected public release includes every documented current feature and potential direction, plus the 24 original requirements. Application source, installers, private histories, credentials, full transfer packages and the original internal pitch remain private.

AI-assisted workflows supported the creation of this project.